RUNGTA INTERNATIONAL SKILLS UNIVERSITY

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SCHOOL OF INFORMATION TECHNOLOGY

BACHELOR OF COMPUTER APPLICATION

<table>
<tr><td>PS No.</td><td>35</td></tr>
<tr><td>Project Title</td><td>Currency Converter</td></tr>
<tr><td>Problem Statement</td><td>Build a console currency converter that converts an amount between supported currencies using exchange rates stored in a dictionary and displays the result rounded to 2 decimal places.
Language: Python 3 Concepts: Functions, Conditions, Operators, Dictionaries
Note: Rates to INR: 1 USD = 83, 1 EUR = 90, 1 GBP = 105. Supported codes: INR, USD, EUR, GBP. Every conversion goes through INR. Results are rounded to 2 decimal places.</td></tr>
<tr><td>Test Cases / Case Statements (4 cases)</td><td>
<table>
<tr><th>INPUT</th><th>EXPECTED OUTPUT</th></tr>
<tr><td colspan="2">Test Case 1: Normal Input</td></tr>
<tr><td>Amount: 166
From: INR
To: USD</td><td>166 INR = 2.00 USD</td></tr>
<tr><td colspan="2">Test Case 2: Boundary / Minimum Input</td></tr>
<tr><td>Amount: 0 (minimum)
From: USD
To: INR</td><td>0 USD = 0.00 INR</td></tr>
<tr><td colspan="2">Test Case 3: Invalid Input</td></tr>
<tr><td>(a) From: JPY
(b) Amount: abc
(c) Amount: -50</td><td>(a) Unsupported currency 'JPY'. Available: INR, USD, EUR, GBP
(b) Invalid amount. Enter a number.
(c) Amount cannot be negative.</td></tr>
<tr><td colspan="2">Test Case 4: Different Valid Scenario</td></tr>
<tr><td>(a) 100 USD to EUR
(b) 2 GBP to INR
(c) 45 EUR to USD</td><td>(a) 100 USD = 92.22 EUR
(b) 2 GBP = 210.00 INR
(c) 45 EUR = 48.80 USD</td></tr>
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</td></tr>
<tr><td>Bare Minimum Requirements</td><td>1. Write the solution in Python 3 (one .py file) and run it from the console.
2. Keep the exchange rates in a dictionary.
3. Write functions such as convert_to_inr() and convert_currency().
4. Use conditions and operators to convert between any two supported currencies.
5. Validate the currency codes and the amount, and round every result to 2 decimal places.</td></tr>
<tr><td>Instruction</td><td>1. Understand the problem statement and define the problem clearly.
2. Develop the solution using proper logic and programming concepts.
3. Implement all required features and pass the given test cases.
4. Test and debug the project to ensure correct output.
5. Prepare the documentation/report and upload the code and report to GitHub.
6. Complete and submit the project within the given deadline.</td></tr>
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